A Randomized Controlled Trial to Evaluate the Effect of Glycemic Control on Renal Transplantation Outcomes

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Context: Outcomes from intensive glycemic control postrenal transplant have not been studied.

Objective: Our objective was to observe the optimal management of hyperglycemia in patients with diabetes or impaired glucose tolerance receiving renal transplantation.

Design, Setting, and Patients: We conducted a randomized controlled trial with patients undergoing renal transplantation randomized to either iv insulin therapy (intensive) or standard sc insulin therapy while the patients were admitted to the hospital.

Interventions: The study consisted of a 3-day postrenal transplant group treated with intensive iv insulin [blood glucose (BG) = 70–110 mg/dl] or a control group treated with sc insulin (BG = 70–180 mg/dl).

Main Outcome Measure: The primary endpoint was delayed graft function (DGF). Secondary endpoints were glycemic control, graft survival, and acute rejection episodes.

Results: A total of 104 patients were screened and randomized to either the intensive or control condition; however, the intention-to-treat analysis set consisted of only the 93 participants (n = 44 intensive, n = 49 control) that underwent a renal transplant. DGF was present in 18% (eight of 44) of the intensive group and 24% (12 of 49) of the control group ($P = 0.46$). The occurrence of severe hypoglycemia (BG $<$ 40 mg/dl) and severe hyperglycemia (BG $>$ 350 mg/dl) were the primary safety outcome measures. There were nine participants with hypoglycemia identified, seven of which (78%) were in the intensive treatment group ($P = 0.08$). There were 30 instances of hyperglycemia with five participants (11%) in the intensive group and 12 participants (24%) in the control group having at least one hyperglycemic event ($P = 0.10$). For the 11 rejection episodes, nine were in the intensive treatment group ($P = 0.013$).

Conclusions: The primary outcome measure of DGF was not statistically different for the two treatment groups. Regarding longer-term rejection and graft survival, the intensively treated participants were at higher risk for a rejection episode. **(*J Clin Endocrinol Metab* 97: 4399–4406, 2012)**

Diabetes mellitus (DM) is an overwhelming U.S. public health issue; currently, 25.8 million Americans are affected and 79 million American adults have prediabetes (1). The incidence of renal involvement in a patient with DM is about 40%, and DM is the leading cause of end-stage renal disease in the U.S. population (2).

Of all kidney transplant recipients, about 25% have preexisting DM, a 33% increase since 1992 (3). Of the

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Abbreviations: BG, Blood glucose; DGF, delayed graft function; DM, diabetes mellitus; OR, operating room; PACU, postoperative acute care unit.

patients who did not have preexisting DM at the time of kidney transplantation, 20–50% developed significant glucose intolerance afterward due to stress hyperglycemia, steroid use, immunosuppressive agents, and other medications (4–7). Therefore, glycemic control after transplantation is as important to patients with DM as in other settings.

Indicators of kidney transplantation success are based upon graft and patient survival (8). The rate of 1- and 5-yr graft survival after cadaveric renal transplantation is now around 90 and 70%, respectively. The primary decrease in graft and recipient survival rates is seen during the first 12 months after transplantation (9, 10). Historically, delayed graft function (DGF) has been defined by the need for dialysis within the first week after renal transplantation (8). Most centers report a DGF rate of 20–40%, but large variations exist in reporting (5–93%) based on the graft source (living *vs.* cadaveric donor), graft procurement, and preservation including cold ischemia time (11–13). Episodes of acute kidney rejection are more prevalent in the presence of DGF (8).

Experimental evidence supports a cause and effect relationship between acute hyperglycemia and poor clinical outcomes. Factors thought to contribute to these concepts are leukocyte action, Ig production, wound healing, collagen production, endothelial function, cardiac performance, and fluid balance (14, 15). Also, insulin may have beneficial vascular and antiinflammatory effects (16). Lowering the plasma glucose with insulin results in a reduction of free fatty acids and stimulates nitric oxide synthase, which results in an elevation of nitric oxide (17, 18). Other circulating inflammatory markers improved with plasma glucose lowering with insulin are IL-6, TNF-α, C-reactive protein, fibrinogen, and others (19). Therefore, it is thought that good glycemic control leads to improved metabolic outcomes in acutely ill patients.

Umpierrez *et al.* (20) have shown that patients on general medical and surgical wards with new and preexisting hyperglycemia had an 18-fold increase in hospital mortality, and patients with known diabetes had a 2.7-fold increase in hospital mortality compared with normoglycemic subjects. Oomichi *et al.* (21) investigated the effect of glycemic control on the mortality of 114 patients with diabetes and chronic kidney disease who were undergoing hemodialysis. Results indicated that good glycemic control (glycated hemoglobin <6.5%) was an independent predictor of improved prognosis. Thus, good glucose control is beneficial; it reduces mortality, nosocomial infections, and stroke events.

Subjects and Methods

This study was a randomized, unblinded, controlled clinical trial that was approved by the Institutional Review Board at the Medical University of South Carolina (MUSC). All patients provided written consent. All patients were managed by the Diabetes Management Service (DMS) at MUSC, who followed the assigned protocols for glycemic control. After discharge from the hospital, the patients were followed up by the DMS team.

TABLE 1. Insulin treatment protocols

Protocols
- A. Control group treated with insulin NPH or glargine and aspart
  - Discontinue all antidiabetic drugs on admission
  - Check FSBG every hour while in the OR and treat with rapid-acting (sc) insulin as needed to aim for target BG levels (70–180 mg/dl)
  - Check FSBG every 4 h while in the recovery room (PACU); cover with NPH or glargine and aspart insulin to aim for target BG levels at 70–180 mg/dl
  - After transfer to the transplant unit while NPO, check FSBG every 4 h and treat with NPH or glargine and aspart (sc) insulin to maintain glycemic control at 70–180 mg/dl
  - After diet consumption, BG levels are checked before meals, at bedtime, and at 0300 h; cover BG with NPH/glargine and aspart tid ac, hs, and 0300 h to maintain BG at 70–180 mg/dl
  - Treat for hypoglycemia if FSBG is <70 mg/dl, same as hypoglycemia standing order
  - Obtain serum glucose if FSBG is <40 mg/dl or >350 mg/dl
  - All BG levels will be checked by FSBG method using the Precision QID monitor
  - At discharge, place on a regimen of a minimum of one to two insulin (sc) injections per day to maintain glycemic control at 90–180 mg/dl
- B. Intensive group treated with iv regular insulin
  - Discontinue oral antidiabetic drugs on admission
  - FSBG now and every 1 h
  - Start iv 250 U insulin regular/250 ml 0.9% NaCl
  - Hang insulin, flush line with and waste 50 ml of insulin solution for initial bag and all iv tubing changes
  - Start insulin infusion on pump at rate determined by the iv insulin infusion calculator for target BG levels of 70–110 mg/dl
  - Use MUSC Web-based iv insulin infusion calculator to change the rate as ordered:
    - (BG − 60) × multiplier = rate of insulin infusion per hour
    - The insulin dose must be recalculated every time FSBG is checked with the latest result, even if the multiplier does not change
  - Treat for hypoglycemia if FSBG is <70 mg/dl as directed by the iv insulin infusion calculator, same as hypoglycemia standing order
  - Obtain serum glucose if FSBG is <40 mg/dl or >350 mg/dl
  - Call the service resident/fellow if:
    - FSBG > 200 × 4 after 4 h of iv insulin
    - Patient requires greater than 25 U/h insulin
    - FSBG is below 70 mg/dl twice in a row
    - If the patient is eating by mouth, the resident/fellow will write for sc insulin to cover carbohydrate intake
  - At discharge, place patient on a regimen of multiple insulin (sc) injections per day to maintain glycemic control at 70–140 mg/dl

NPH, Neutral protamine hagedorn; FSBG; fingerstick blood glucose; NPO, nothing by mouth; TID AC, three times per day before meals; hs, bedtime.

This study included renal transplant candidates admitted to MUSC who were 18 yr of age or greater and who had a DM diagnosis (type 1 and type 2), a fasting blood glucose (BG) over 100 mg/dl per admission screening labs, and a random BG over 120 mg/dl per admission screening labs. These subjects were willing and able to provide informed consent and were awaiting a living or cadaveric kidney transplant. Exclusion criteria included patients with a history of an active gastrointestinal bleed 3 months previously, patients who were scheduled to receive a simultaneous pancreas transplant, patients with a history of a functioning pancreatic transplant, patients currently managed on an insulin pump, patients who were unable or unwilling to provide informed consent, and patients who were unable to commit to the study protocol, including the outpatient follow-up phase of care.

Patients were randomized to either the intensive group with iv insulin or the control group with sc insulin (Table 1). The study consisted of a 3-d post-renal-transplant group treated with intensive iv insulin (BG 70–110 mg/dl) or a control group treated with sc insulin (BG 70–180 mg/dl). After randomization, all subjects underwent laboratory studies for primary analysis. All patients remained on call to the operating room (OR) for surgery. The research protocol was implemented upon arrival to the OR. After randomization, additional laboratory studies (glycated hemoglobin, hemoglobin and hematocrit, lipids, and comprehensive metabolic panel) were obtained. Non-routine-care labs such as C-peptide values were obtained at study entry and 3 months. All study subjects were managed according to the assigned treatment group and received routine nursing care and patient education specific to self-care management (insulin therapy and BG monitoring) by the transplant unit Registered Nurses and assigned Certified Diabetes Educators.

Patients were randomly assigned to one of two treatment regimens (Table 1). A control group consisted of BG levels checked every hour while in the OR via indwelling venous cannula or finger-stick BG and was treated with rapid-acting sc insulin as needed to aim for target BG levels (70–180 mg/dl). Control subjects had BG levels checked every 4 h while in the postoperative acute care unit (PACU), treated with long-acting and rapid-acting sc insulin to aim for target BG levels (70–180 mg/dl), and were followed by the DMS team. After control subjects were transferred to the transplant unit, BG levels were checked every 4 h and treated with long-acting (neutral protamine hagedorn insulin or glargine insulin) and rapid-acting (aspart) sc insulin to maintain glycemic control at 70–180 mg/dl and were followed by the DMS team. After diet consumption, BG levels were checked before meals, at bedtime, and at 0300 h. The doses of insulin were adjusted daily according to protocol (Table 1). It was DMS protocol to maintain glycemic control at 70–180 mg/dl for the remainder of the at least 72 h. Control subjects were placed on a regimen of a minimum of one to two insulin sc injections per protocol at discharge to maintain glycemic control at 90–180 mg/dl.

An experimental group that consisted of BG level checked in the OR via indwelling venous cannula or finger-stick BG just before starting the iv insulin infusion (Table 1). The iv insulin infusion solution was prepared as a mixture of 250 U regular insulin (Novolin Regular) with 250 mg 0.9% NaCl rendering 1 U insulin/ml saline. The formula was (current BG − 60 mg/dl) × 0.03 = rate of insulin infusion per hour. BG levels were maintained according to study protocol (70–110 mg/dl). BG levels were checked every 1–2 h per study protocol to maintain glycemic control at 70–110 mg/dl and were followed by the DMS team. The at least 72-h time period started at the beginning of surgery and continued until 0700 h on postoperative day 3. Experimental subjects in the intensive group had all BG levels checked by the fingerstick BG method using the Precision QID monitor. BG management followed the iv insulin infusion calculator study protocol (70–110 mg/dl). After diet consumption, BG levels followed the iv insulin infusion calculator study protocol for the remainder of the at least 72 h. The doses of insulin were adjusted daily according to protocol (Table 1). After the 72-h period concluded, the intensive group was transitioned to long-acting and rapid-acting sc insulin for glycemic control (70–140 mg/dl).

All patients who met the inclusion criteria for this study received induction with an IL-2 receptor antagonist (dacliuzumab or basiliximab 1 mg/kg or 20 mg iv, respectively), just before graft reperfusion and on postoperative d 7, or with an anti-T-depleting agent (Thymoglobulin iv 1.5 mg/kg for a total of three to five doses daily administered). Maintenance immunosuppression consisted of tacrolimus tablets, adjusted to maintain target trough whole blood concentration from 8–15 ng/ml for wk 1–6 after transplant, 6–12 ng/ml for wk 7–12, and 5–10 ng/ml after wk 12. In addition, all the patients received mycophenolate mofetil (CellCept 1 g by mouth twice a day or Myfortic 720 mg by mouth twice a day), and prednisone, tapered to 5 mg/d by 3 months after transplant. All the biopsy-proven rejection episodes were graded according to the Banff classification score system (22–24). Acute cellular rejection was treated with methylprednisolone 500 mg boluses (three to five doses) and/or Thymoglobulin (1.5 mg iv for five to 10 courses). In case of antibody-mediated rejection, patients also received Igs iv and plasmapheresis sessions according to the clinical response.

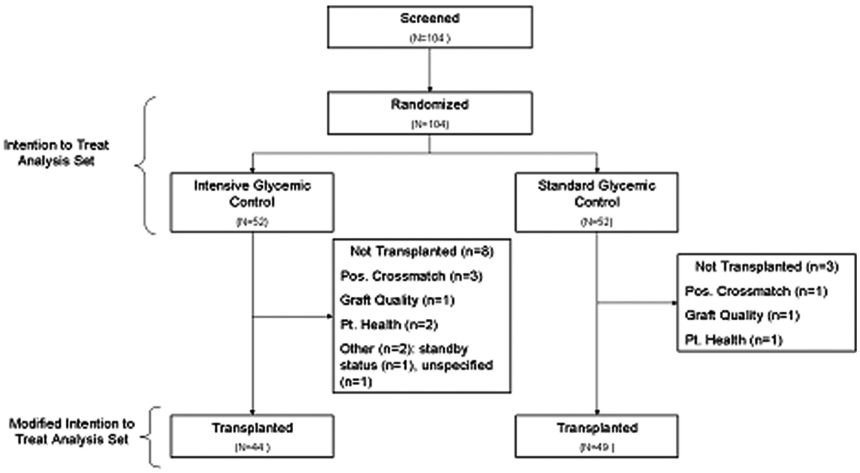

FIG. 1. Subject flow and disposition diagram. Pos., Positive; Pt, patient.

Statistical considerations

Sample size estimation

This study was designed as an early-phase clinical trial to provide preliminary evidence for the design and implementation

of a multicenter clinical trial. For the present study, the statistical error rates were established *a priori* at 0.15 and 0.20 for the type I and type II error rates, respectively. The sample size for the clinical trial was based on feasibility of enrollment along with power to detect a clinically relevant effect size in secondary endpoints (*e.g.* BG levels and biomarkers) at the $P < 0.15$ level of significance. The upper bound for the recruitment potential of this single clinical site was based on transplant volume. The transplant center estimated approximately 150 renal transplants per year; of these, it was conservatively estimated that approximately 30% of these patients would be diabetic, eligible, and willing to participate in this study. Thus, approximately 45 patients per year would be the expected accrual annual rate. This rate for a projected accrual period of 27 months would be expected to yield approximately 90 participants in the trial. Using the feasible sample size of 90 participants, there was 80% power to detect an effect size of 0.48 for the secondary outcomes (*i.e.* a 0.48-SD difference in continuous outcomes).

Primary analysis

The surrogate endpoint of DGF was chosen as the primary outcome variable for this study to account for the limited follow-up period, the small sample size, and the relatively high graft survival rate. DGF was defined as either a d-10 serum creatinine value over 2.5 mg/dl or the need for dialysis within the first 7 d after transplant. To test the null hypothesis that the probability of DGF was equal between the two treatment groups, a χ^2 test was used. Time for graft failure or biopsy-confirmed acute rejection was analyzed using a Kaplan-Meier product limit estimator and the log-rank test. The BG measurements were summarized according to the methodology given by Goldberg *et al.* (25). To compare mean BG readings between the intensive and control conditions, hierarchical linear models were used to account for the clustering of BG readings within participants. A planned interim analysis for efficacy was conducted when 50% of the subjects had been randomized, and the last randomized subject had been followed for at least 2 wk (to measure DGF). The actual interim analysis was conducted with an information fraction of 57% (51 of 90 participants). To control the type I error rate at time of interim analysis, the O'Brien-Fleming-like α-spending function was used. Analyses were conducted in accordance with intention-to-treat principles. However, a modified version was necessary for this trial. To prepare for necessary iv insulin management, participants had to be randomized before the kidney's arrival to the transplant center. In 11 cases, the transplant was not attempted. As such, these participants were excluded from the analysis. Analyses were conducted using The SAS system version 9.2 (SAS Institute, Cary, NC).

Results

Sample description

A total of 104 people were screened for eligibility and were randomized to either the intensive or the control

TABLE 2. Donor and recipient characteristics

Variable	Intensive treatment (n = 44)		Standard treatment (n = 49)		*P* value
	Mean or n	SD or %	Mean or n	SD or %	
Recipient demographics and history					
Age at time of transplant (yr)	58	9.8	56.3	9.6	0.35
Sex, male	30	68	35	71	0.73
Race					
African-American	28	64	28	57	0.60
White	15	34	18	37	
Other	1	2	3	6	
Diabetes	43	98	49	100	0.47
Impaired fasting glucose	1	2	0	0	
Type 1	6	14	3	6	
Type 2	37	84	46	94	
Glycated hemoglobin	44	7.0 (1.8)	49	7.2 (1.7)	0.65
Hyperlipidemia	20	45	24	49	0.73
Years of dialysis	3.1	1.9	2.8	1.9	0.44
Hypertension	42	95	49	100	0.13
Cardiovascular disease	17	39	16	33	0.55
Previous myocardial infarction	5	11	5	10	0.86
Congestive heart failure	4	9	7	14	0.44
Atrial fibrillation	4	9	6	12	0.62
Cerebrovascular events	5	11	6	12	0.90
Vascular disease	11	25	15	31	0.55
Donor and graft history					
Age	40.9	16.6	40.2	15.0	0.76
Diabetes	2	5	5	11	0.28
Cadaveric donor	35	80	38	78	0.82
Hypertension	10	0	10	0	0.86
Coronary artery disease	1	2	2	4	0.60
Smoking	10	24	15	33	0.33
Cerebrovascular event	14	33	16	36	0.83
Cold ischemic time <20 h	26	60	29	59	0.90

condition. After the initial enrollment, the groups were equally balanced with 52 participants per treatment group; however, the intention-to-treat analysis set consisted of only the 93 participants (n = 44 intensive, n = 49 control) who underwent renal transplantation.

The reasons the 11 participants were not transplanted are identified in Fig. 1.

The demographic, medical history, and peritransplant summary of the 93 participants, included in the intention-to-treat analysis, is presented in Table 2. The majority of the participants were African-American (60%, 56 of 93) and male (70%, 65 of 93) and had type 2 diabetes (89%, 83 of 93). Cadaveric donors supplied the majority of the kidneys (78%, 73 of 93), and the cold ischemic time was less than 20 h for 60% of the organs. The two treatment groups were well balanced across the potential confounding variables, including donor kidney status. Postoperative follow-up was based on standard of care and was not influenced by randomization.

Primary endpoint

Delayed graft function

An interim analysis was conducted with 51 participants. With an expected sample size of 90 participants, 51 participants represented an information fraction of 57% (51 of 90). The interim stopping boundary was $Z > 1.98634$, or a χ^2 value of 3.9455, based on the O'Brien-Fleming-like α-spending function (57% information fraction, $\alpha = 0.15$ two sided). At the time of interim analysis, the response proportions in the intensive and control conditions were 14% (three of 22) and 38% (11 of 29) ($P = 0.0542$, $\chi^2 = 3.7075$), respectively. Because the level of significance did not exceed the *a priori* stopping boundary, the study Data and Safety Monitoring Board recommended continuation of the trial. Figure 2 plots the natural logarithm of the odds ratio for DGF as a function of accrued data in the trial. Early in the course of the study, there was significant variability in the estimated treatment effect (as expected), but the indication of a treatment effect at time of interim analysis did not persist to the end of the study. In particular, the group differences between the proportion of participants and DGF decreased. The final response rates were 18% (eight of 44) in the intensive group, and 24% (12 of 49) in the control group ($P = 0.46$), which did not reach the *a priori* level of significance (*e.g.* $P < 0.15$). Similarly, the serum creatinine values did not differ between groups at d 10 [intensive *vs.* standard control, mean (SD) = 2.3 (2.0) *vs.* 2.4 (2.1); $P = 0.8$].

Secondary endpoints

Glycemic control

Patients treated with iv insulin demonstrated greater improvement in glycemic control than those treated with sc insulin ($P < 0.001$) (Fig. 3). The occurrence of severe hypoglycemia (BG < 40 mg/dl) and severe hyperglycemia (BG > 350 mg/dl) were the primary safety outcome measures. A total of 4424 BG readings were obtained during the inpatient phase of the study. There were 10 events of hypoglycemia identified, eight (80%) of which were in the intensive treatment group. One participant in the intensive group had two occurrences of hypoglycemia, so the total number of patients with hypoglycemia was seven (16%) in the intensive group and two (4%) in the control condition ($P = 0.08$). Hyperglycemia, however, tended to be higher in the control condition. There were 30 instances of hyperglycemia across the participants, with 5 participants (11%) in the intensive group and 12 participants (24%) in

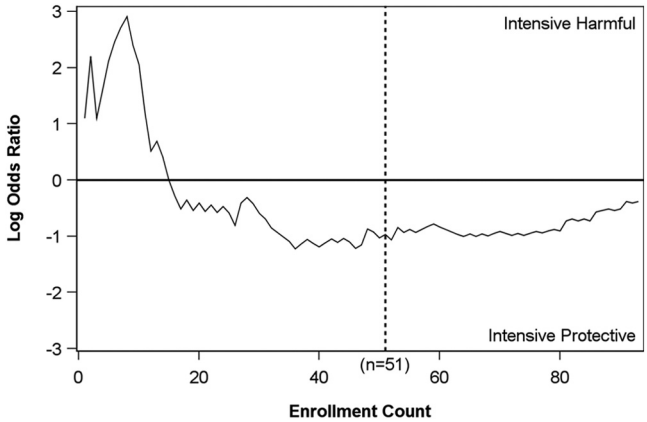

FIG. 2. Treatment effect as data accrued. The treatment effect is quantified by the log-odds ratio (base e) for the development of DGF of the intensively treated participants relative to the control participants. The *horizontal reference line* represents a log-odds ratio of 0 (odds ratio = 1.0). The *area above the horizontal reference line* is consistent with odds ratios indicating increased odds of DGF in the intensively treated participants. The *area below the horizontal reference line* is consistent with the intensively treated patients being protected from DGF. The *vertical dashed line* represents the treatment effect at time of interim analysis. The Data and Safety Monitoring Board recommended continuation of the study, and the trial ended with a log-odds ratio of −0.37 (odds ratio = 0.69; $P = 0.46$).

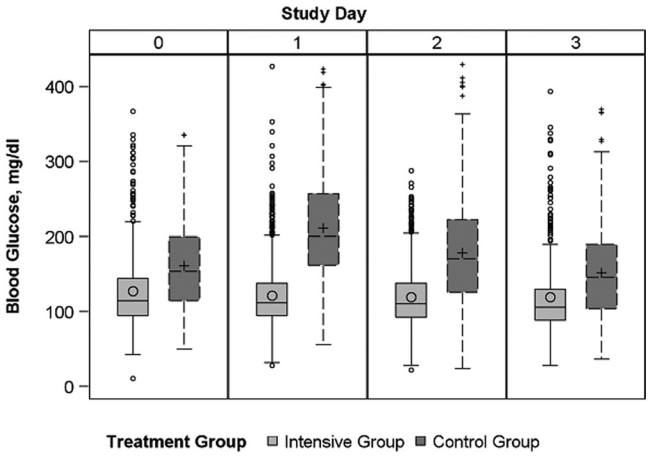

FIG. 3. Changes in BG concentrations in patients treated with iv insulin (intensive group) *vs.* insulin (control group).

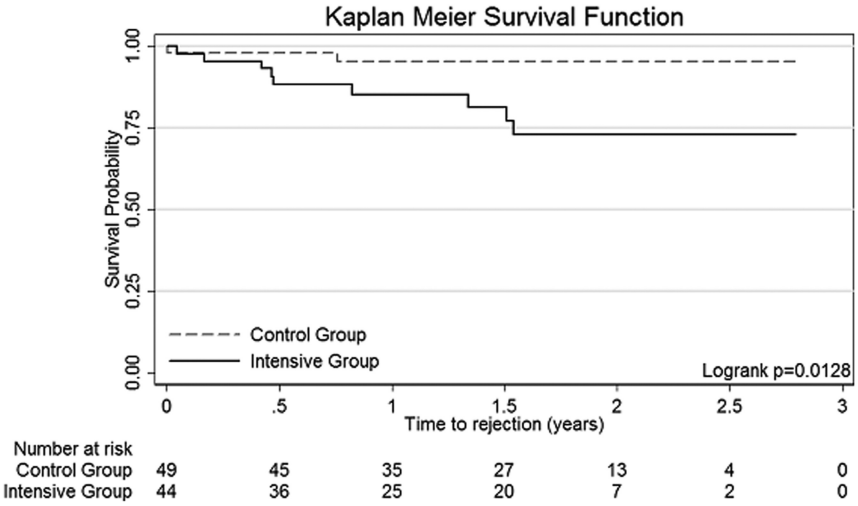

FIG. 4. Product-limit (Kaplan-Meier) survival curves modeling the time to nephrectomy or biopsy-confirmed rejection. The hazards between the two treatment groups differ statistically based on the log-rank statistic ($P = 0.0128$).

the control group, having at least one hyperglycemic event ($P = 0.10$). The mean BG readings, however, were lower in the intensively treated group. The group-specific mean (SE) BG level during the inpatient stay for the intensively treated participants was 122.5 mg/dl (3.4), which was statistically lower than the 177.3 mg/dl (3.4) for the control group ($P < 0.001$). These findings were heavily weighted toward the more frequent assessment during the day of surgery and up to 3 d after surgery. After postoperative d 3, there were only 121 BG readings across 80 patients. During this time, BG did not differ by study group [$P = 0.9$; estimated means (SE) were 156.7 (9.7) and 155.0 (8.9) for intensive and control subjects, respectively]. Data on BG levels after the inpatient period of that study were not captured.

Graft loss and graft survival

Rejection of the graft was determined by biopsy according to the Banff score, classified as acute cellular rejection, antibody-mediated rejection, or removal of the kidney. The median follow-up time from surgery, either September 1, 2010 (close of data collection), or the occurrence of a rejection episode (whichever came first), was 1.5 yr. There were 11 rejection events (11.8%), including three nephrectomies. One graft was removed on postoperative d 1 due to a thrombosis. This participant was randomized to the control group.

The second graft was removed after 1.7 yr, and the remaining nephrectomy case occurred 2.7 yr after transplant because of progressive graft failure. Of note, this last nephrectomy case occurred over a year after the initial abnormal biopsy result. The latter two cases of nephrectomy were both randomized to the intensive control group. For the 11 rejection episodes, nine were in the intensive treatment group, and the intensively treated participants had a higher hazard (risk) for developing a rejection. All rejection episodes occurred in patients that did not experience a severe hypoglycemic event. Two of the rejections, both on intensive control, had type 1 diabetes. Figure 4 plots the survival curves for the two treatment groups.

Discussion

This is the first prospective randomized clinical trial aimed to compare the effects of blood sugar control, using iv insulin (BG target 70–110 mg/dl) or sc insulin (BG target 70–180 mg/dl) in patients with impaired glucose tolerance or DM undergoing renal transplantation. Although there was a difference in the point estimates in the DGF endpoint favoring intensive therapy, the difference (18 *vs.* 24%) did not reach statistical significance. Participants on intensive therapy experienced increased risk for rejection episodes and hypoglycemic events. Overall, for kidney-pancreas transplants, the DGF rate is a very rare situation, much less than for kidney transplant patients alone. This is mainly due to the excellent selection of donors for kidney-pancreas (age and no obesity). However, the incidence of early rejection for kidney-pancreas transplants is comparable to kidney transplant alone.

There were differences in glycemic control between the two treatment groups. As expected, there were more hypoglycemic events in the intensively managed glycemic groups. It is speculated that the increased events of hypoglycemia in the intensively treated group may have contributed to the increased rejection event rate (26); however, in this study, none of the cases of rejection was associated with hypoglycemia events. Glucose control in the renal transplant patient is challenging. Due to immunosuppressive agents and corticosteroid dosing, insulin dosing is a dynamic process. All patients receiving kidney transplantation may develop hyperglycemia due to corticosteroid dosing and other immunosuppressive agents given in the operating room and during the posttransplantation period (27). The means by which hyperglycemia exacerbates ischemia/reperfusion injury is not known but may be due to inflammation with an increased incidence of macrophages and heightened oxidative stress (27).

We recognize the following limitations in the study. This was a small, single-site open clinical trial that enrolled

93 patients. The clinical trial was designed as a pilot study and was not powered to detect small risk differences in binomial outcomes, *e.g.* DGF. The study observed a 26% reduction in risk for DGF (relative risk = 0.74) in the intensive group, but the small sample size produced a CI range that was neither conclusive nor highly informative (95% CI = 0.33–1.65). Thus, although a type II error could have occurred due to limited power, the failure to reject the null hypothesis for the primary endpoint, coupled with the observed increase in risk for the rejection episodes and hypoglycemia risk, attenuates the concerns regarding potential false-negative results. These concerns are amplified by the findings that intensive glycemic control may be associated with other risks (28). However, caution must be exercised in interpreting the observed effect because it may be estimated with poor precision (29). Biomarker data and stored specimens have been collected. The interactions of the intensive therapy on these biomarkers and subsequent risks of events are currently being investigated.

A limitation of the trial was that the majority of the patients enrolled in the study were African-American, and therefore the findings cannot be extended to groups of other ethnic groups. Another limitation of the trial was that staff and study personnel were not blinded as to the patient's treatment group assignments. A larger multicenter study is needed to provide more definitive evidence, but based on the findings of this study, clinical equipoise may be questionable. Should additional clinical trials be considered, stopping rules for acute rejection may be necessary, particularly because the findings in this trial support the outcome of the NICE-SUGAR (The Normoglycemia in Intensive Care Evaluation-Survival Using Glucose Algorithm Regulation) study in that intensive glucose control may not benefit patients undergoing renal transplantation and may be harmful for kidney rejection (28).

In conclusion, our trial showed that a BG target of less than 180 mg/dl resulted in a lower incidence of kidney rejection than a BG target of 70–110 mg/dl. Due to these results, intensive glycemic control at the time of kidney transplantation is not recommended. Further study of underlying mechanisms that may explain these findings is recommended.

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K.L.H. and R.E.C. designed the study, conducted all analyses, interpreted data, prepared and reviewed the manuscript, had full access to all data in the study, and takes full responsibility for the integrity of data and the accuracy of data analysis. M.F.E. interpreted data, provided guidance for sophisticated analysis, and wrote, reviewed, and edited the manuscript. P.B., A.L., and S.B. reviewed and edited the manuscript. N.J.F. assisted with design of the study. L.K. interpreted data and prepared and reviewed the manuscript.

Disclosure Summary: K.L.H. is conducting clinical research trials with Eli Lilly, Sanofi aventis, and Novo Nordisk. K.L.H. is on the Speaker's Bureau for Eli Lilly, Sanofi aventis, Boehringer Ingelheim, and Amylin Pharmaceuticals. There are no other known potential conflicts of interest.

References

1. **Centers for Disease Control and Prevention** 2011 National diabetes fact sheet. Atlanta, GA: U.S. Department of Health and Human Services, Centers for Disease Control and Prevention
2. **de Boer IH, Rue TC, Hall YN, Heagerty PJ, Weiss NS, Himmelfarb J** 2011 Temporal trends in the prevalence of diabetic kidney disease in the United States. JAMA 305:2532–2539
3. **Gaston RS, Alveranga DY, Becker BN, Distant DA, Held PJ, Bragg-Gresham JL, Humar A, Ting A, Wynn JJ, Leichtman AB** 2003 Kidney and pancreas transplantation. Am J Transplant 3(Suppl 4): 64–77
4. **Cosio FG, Pesavento TE, Osei K, Henry ML, Ferguson RM** 2001 Post-transplant diabetes mellitus: increasing incidence in renal allograft recipients transplanted in recent years. Kidney Int 59:732–737
5. **Kasiske BL, Snyder JJ, Gilbertson D, Matas AJ** 2003 Diabetes mellitus after kidney transplantation in the United States. Am J Transplant 3:178–185
6. **Woodward RS, Schnitzler MA, Baty J, Lowell JA, Lopez-Rocafort L, Haider S, Woodworth TG, Brennan DC** 2003 Incidence and cost of new onset diabetes mellitus among U.S. wait-listed and transplanted renal allograft recipients. Am J Transplant 3:590–598
7. **Lane JT, Dagogo-Jack S** 2011 Approach to the patient with new-onset diabetes after transplant (NODAT). J Clin Endocrinal Metab 96:3289–3297
8. **Daly PJ, Power RE, Healy DA, Hickey DP, Fitzpatrick JM, Watson RW** 2005 Delayed graft function: a dilemma in renal transplantation. BJU Int 96:498–501
9. **Grochowiecki T, Szmidt J, Gałazka Z, Nazarewski S, Madej K, Meszaros J, Paczek L, Durlik M, Wyzgał J, Grygiel K, Wojtaszek M, Piwowarska J, Kañski A** 2006 Comparison of 1-year patient and graft survival rates between preemptive and dialysed simultaneous pancreas and kidney transplant recipients. Transplant Proc 38:261–262
10. **Hariharan S, Johnson CP, Bresnahan BA, Taranto SE, McIntosh MJ, Stablein D** 2000 Improved graft survival after renal transplantation in the United States, 1988 to 1996. N Engl J Med 342:605–612

11. **Brennan TV, Freise CE, Fuller TF, Bostrom A, Tomlanovich SJ, Feng S** 2004 Early graft function after living donor kidney transplantation predicts rejection but not outcomes. Am J Transplant 4:971–979
12. **Brook NR, White SA, Waller JR, Veitch PS, Nicholson ML** 2003 Non-heart beating donor kidneys with delayed graft function have superior graft survival compared with conventional heart-beating donor kidneys that develop delayed graft function. Am J Transplant 3:614–618
13. **Koning OH, Ploeg RJ, van Bockel JH, Groenewegen M, van der Woude FJ, Persijn GG, Hermans J** 1997 Risk factors for delayed graft function in cadaveric kidney transplantation: a prospective study of renal function and graft survival after preservation with Univeristy of Wisconsin solution in multi-organ donors. European Multicenter Study Group. Transplantation 63:1620–1628
14. **McMahon MM, Bistrian BR** 1995 Host defenses and susceptibility to infection in patients with diabetes mellitus. Infect Dis Clin North Am 9:1–9
15. **Metchick LN, Petit Jr WA, Inzucchi SE** 2002 Inpatient management of diabetes mellitus. Am J Med 113:317–323
16. **Dandona P, Aljada A, Bandyopadhyay A** 2003 The potential therapeutic role of insulin in acute myocardial infarction in patients admitted to intensive care and in those with unspecified hyperglycemia. Diabetes Care 26:516–519
17. **Steinberg HO, Brechtel G, Johnson A, Fineberg N, Baron AD** 1994 Insulin-mediated skeletal muscle vasodilation is nitric oxide dependent. A novel action of insulin to increase nitric oxide release. J Clin Invest 94:1172–1179
18. **Zeng G, Quon MJ** 1996 Insulin-stimulated production of nitric oxide is inhibited by wortmannin. Direct measurement in vascular endothelial cells. J Clin Invest 98:894–898
19. **Hirsch IB** 2004 Effect of insulin therapy on nonglycemic variables during acute illness. Endocr Pract 10(Suppl 2):63–70
20. **Umpierrez GE, Isaacs SD, Bazargan N, You X, Thaler LM, Kitabchi AE** 2002 Hyperglycemia: an independent marker of in-hospital mortality in patients with undiagnosed diabetes. J Clin Endocrinol Metab 87:978–982
21. **Oomichi T, Emoto M, Tabata T, Morioka T, Tsujimoto Y, Tahara H, Shoji T, Nishizawa Y** 2006 Impact of glycemic control on survival of diabetic patients on chronic regular hemodialysis: a 7-year observational study. Diabetes Care 29:1496–1500
22. **Solez K, Colvin RB, Racusen LC, Haas M, Sis B, Mengel M, Halloran PF, Baldwin W, Banfi G, Collins AB, Cosio F, David DS, Drachenberg C, Einecke G, Fogo AB, Gibson IW, Glotz D, Iskandar SS, Kraus E, Lerut E, Mannon RB, Mihatsch M, Nankivell BJ, Nickeleit V, Papadimitriou JC, et al.** 2008 Banff 07 classification of renal allograft pathology: updates and future directions. Am J Transplant 8:753–760
23. **Sis B, Mengel M, Haas M, Colvin RB, Halloran PF, Racusen LC, Solez K, Baldwin 3rd WM, Bracamonte ER, Broecker V, Cosio F, Demetris AJ, Drachenberg C, Einecke G, Gloor J, Glotz D, Kraus E, Legendre C, Liapis H, Mannon RB, Nankivell BJ, Nickeleit V, Papadimitriou JC, Randhawa P, Regele H, et al.** 2010 Banff '09 meeting report: antibody mediated graft deterioration and implementation of Banff working groups. Am J Transplant 10:464–471
24. **Solez K** 2010 History of the Banff classification of allograft pathology as it approaches its 20th year. Curr Opin Organ Transplant 15:49–51
25. **Goldberg PA, Bozzo JE, Thomas PG, Mesmer MM, Sakharova OV, Radford MJ, Inzucchi SE** 2006 "Glucometrics": assessing the quality of inpatient glucose management. Diabetes Technol Ther 8:560–569
26. **Pomposelli JJ, Baxter 3rd JK, Babineau TJ, Pomfret EA, Driscoll DF, Forse RA, Bistrian BR** 1998 Early postoperative glucose control predicts nosocomial infection rate in diabetic patients. JPEN J Parenter Enteral Nutr 22:77–81
27. **Parekh J, Bostrom A, Feng S** 2010 Diabetes mellitus: a risk factor for delayed graft function after deceased donor kidney transplantation. Am J Transplant 10:298–303
28. **NICE SUGAR Study Investigators; Finfer S, Chittock DR, Su SY, Blair D, Foster D, Dhingra V, Bellomo R, Cook D, Dodek P, Henderson WR, Hébert PC, Heritier S, Heyland DK, McArthur C, McDonald E, Mitchell I, Myburgh JA, Norton R, Potter J, Robinson BG, Ronco JJ** 2009 Intensive versus conventional glucose control in critically ill patients. N Engl J Med 360:1283–1297
29. **Kraemer HC, Mintz J, Noda A, Tinklenberg J, Yesavage JA** 2006 Caution regarding the use of pilot studies to guide power calculations for study proposals. Arch Gen Psychiatry 63:484–489